

Credit Lines for Pseudonymous Borrowers: Identity-Gated Bayesian Lending under an On-Chain Issuance Budget

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Abstract

A lending pool on a public blockchain lends without collateral against credit lines that an issuer publishes for pseudonymous accounts. The pool's contract bounds what the issuer can put at risk: a budget on the sum of lines, charged on each account's highest line since the line was last unused, and a cap on how much one report may add. The issuer must then decide whom to lend to and how much, from thin and manipulable evidence. We give a policy that does so in four steps. Only accounts with a costly identity are eligible, and a tier's cap never exceeds the cost of a fraudulent identity in it, which makes buying identities unprofitable whatever the attacker does. Repayment history enters a Beta-Binomial estimate of the probability of default only in proportion to the risk lenders actually bore, so every route to evidence that costs nothing earns nothing. Each line maximises expected profit under exponential demand, at $\mu \ln(i/p)$. Lines are allocated within the budget and the per-report cap by a greedy rule that is optimal because the constraints form a laminar matroid, and every report is planned to land even if an adversary releases held budget first. In a 24-month simulation the contract accepted all 925 reports, predicted default rates matched realised ones in every decile, lines grew for low-risk borrowers and shrank for high-risk ones, and attackers buying identities lost money in every tier unless identities cost far less than the issuer assumed.

Keywords: OR in banking, credit scoring, Bayesian estimation, resource allocation, matroids, incentive compatibility

1. Introduction

Credit scoring decides who receives credit and how much (Thomas et al., 2017). The operational research literature has studied it as classification (Lessmann et al., 2015) and as a decision problem in which the lender sets limits to maximise profit (Crook et al., 2007). This paper studies a setting that changes three of the standing assumptions. Borrowers are pseudonymous accounts on a public blockchain, so an attacker can create as many as it likes (Douceur, 2002). Their history is public and partly manufacturable: a loan repaid within a day pays no interest, and capital returned at the end of a history can be reused for the next account. And the lender, here called the issuer, does not lend itself. It publishes lines into a pool funded by others, whose contract limits how much the issuer can put at risk.

The pool's contract charges an issuance budget on each account's highest line since the line was last unused, releases that budget only when the account has no open loans and no commitments,

¹Claude Code is an AI agent made by Anthropic, working with the Microcredit Protocol Project; the research was directed by the project's human operator. Sources, data and code: <https://github.com/scottonchain/microcredit-theory>.

and caps how much one report may raise the total (Claude Code, 2026). Under those rules, the lenders’ potential loss on the issuer’s lines never exceeds the budget, even if the issuer is compromised. Within that bound, the issuer still has to be accurate and resistant to manipulation: it is a delegated monitor in the sense of Diamond (1984), and its lines are where the pool’s losses come from.

We give a reference policy for this issuer and analyse its properties. The contributions are:

- *An incentive-compatible eligibility rule* (Proposition 1). A tier’s line cap must not exceed the cost of an identity in that tier. Then no number of bought identities, with any history, yields a profit, because the loss an account can cause over its lifetime is at most its highest line plus the dues it paid (Lemma 1).
- *Evidence weighted by risk borne* (Section 4). A repayment counts in a Beta-Binomial posterior only in proportion to the unsecured principal, the time at risk and its recency. Every zero-cost route to evidence has weight zero.
- *A closed-form line* (Proposition 2): with exponential demand, expected profit is concave in the line and maximised at $\mu \ln(i/p)$.
- *Optimal allocation within the contract’s rules* (Proposition 3 and Lemma 2). The feasible raises form a laminar matroid, so a greedy rule is optimal; and the plan respects every rule even when an adversary front-runs it.
- *A simulation* of 24 monthly reports with honest borrowers, identity buyers, a farm of recycled histories and an adversary who front-runs every report.

2. The Setting

The pool. Accounts borrow stablecoins for 30-day terms. An account’s own credit is its issued line plus the dues it has paid, a share of its interest held in a first-loss reserve; other accounts may also back it with credit of their own (Claude Code, 2026). A loan defaults 30 days after its due date and is then written off.

The issuer’s interface. The issuer publishes scores $q(a) \in [0, 1]$; the line is $\ell(a) = M q(a)$ for a maximum line M . The contract keeps a held budget $\beta(a)$ per account. A report that raises $q(a)$ above $\beta(a)$ raises $\beta(a)$ to it, and a report is rejected if it raises $\sum_a \beta(a)$ above a budget B or by more than a per-report cap Δ . Anyone may call a release that lowers $\beta(a)$ to $q(a)$, but only while a has no open loans and no backing commitments. Scores read as zero if no report arrives within a maximum age, so the issuer sends empty reports as heartbeats. Reports carry increasing epochs and at most 500 accounts each.

What the issuer reads. For each account: whether an identity provider has verified it, whether it has ever defaulted, its own loans (principal, term, disbursement, repayment and closing times) and, per loan, the secured backing it held. From its own registry: the identity behind the account and the identity’s tier. It never reads who backs whom. A function of the backing graph is exactly what Sybil accounts can manipulate (Cheng and Friedman, 2005).

3. Eligibility and Incentive Compatibility

An account receives a line only if it holds a verified identity of tier t , obtained at a cost k_t (the price of a fraudulent identity in that tier, plus any expected penalty), uses the identity’s one live address, and has no default on any address of the identity. Everything else receives zero.

Lemma 1 (Lifetime loss per account). *For any account a and any sequence of operations, the losses lenders bear that are attributable to a , namely the credit charged to a as a backer plus the residual of its own defaults, are at most $\hat{\ell}(a) + d(a)$, where $\hat{\ell}(a)$ is the highest line a has ever held and $d(a)$ its dues.*

Proof. The per-account invariant of Claude Code (2026, Lemma 5.1) states that residual exposure plus committed credit plus attributed loss is at most the line plus dues, and each operation of the pool preserves it. With the line replaced by its running maximum $\hat{\ell}(a)$, every step of that proof goes through unchanged: borrowing and committing are bounded by granted credit, which is at most $\hat{\ell}(a) + d(a)$ less the credit already charged; a charge lowers commitments by at least what it adds to the loss; a default moves the residual from exposure into the loss; and lowering a line leaves $\hat{\ell}(a)$ unchanged. Since the left side of the invariant includes the attributed loss, the claim follows. $\square$

Proposition 1 (Incentive compatibility). *Suppose an attacker buys n_t identities in each tier t at cost k_t , then follows any strategy, possibly also holding a share s of the pool as a lender. Its net profit is at most*

$$\sum_t n_t (\min\{\text{cap}_t, M\} - k_t) + I(r - 1 + s(1 - f - r)) + H,$$

where I is the interest its accounts paid, r and f the reserve and fee shares of interest, and H the backing honest accounts committed to it. If $\text{cap}_t \leq k_t$ in every tier, the first two terms are at most zero.

Proof. By Lemma 1, each identity yields at most its highest line, which the policy keeps at or below $\min\{\text{cap}_t, M\}$, plus its dues $d = rI$ in total. The attacker paid k_t per identity and interest I , and recovers through its shares at most $s(1 - f - r)I$ (Claude Code, 2026, Proposition 7.4). The second term is at most $I(r - 1 + 1 - f - r) = -fI \leq 0$. Honest backing is the honest backers’ own risk, charged to their credit. $\square$

Under the condition, a profit-seeking attacker gains nothing from any number of identities through any history; this is the identity-cost form of the result of Friedman and Resnick (2001) that, with cheap pseudonyms, newcomers must pay dues, and it is why identities issued once per person matter (Borge et al., 2017). An attacker that does not seek profit can still cost lenders up to $\sum_t n_t \text{cap}_t$, which under the condition is at most what it spent on identities, and at any moment at most MB on all of the issuer’s lines. Table 1 lists the tiers used below; their costs are illustrative inputs, which an issuer must set from the market price of identities and revisit.

4. Evidence and the Probability of Default

Each closed loan of the account itself is a fractional Bernoulli trial. A loan repaid on time is a good trial with weight

$$w_{\text{good}} = 2^{-\text{age}/h} \cdot \min\left\{1, \frac{\text{principal} - \text{secured}}{P_{\text{ref}}}\right\} \cdot \min\left\{1, \frac{\min\{\text{elapsed}, \text{term}\}}{30 \text{ days}}\right\},$$

Table 1: Tiers: identity cost, line cap, the prior’s strength in trials and its annual PD, a new identity’s line, and the line after 24 clean 30-day trials (USDC).

Tier	Cost k_t	Cap	Prior strength	Prior PD	Starting line	After 24 trials
basic	15	15	57	6.9%	6.1	15.0
standard	40	40	60	5.0%	21.9	38.6
enhanced	120	100	94	3.3%	43.3	54.7

the product of recency with half-life h , unsecured size up to a reference principal P_{ref} , and time at risk; and $w_{\text{good}} = 0$ if the loan closed within the 24-hour window in which no interest accrues. A loan repaid after its due date is a bad trial whose weight is not discounted for stake, so stake cannot hide lateness. Open loans are not yet evidence, and a default ends the identity. Loans of other accounts, including accounts this one backed, are never read.

Let G and B be the sums of good and bad weights. With a $\text{Beta}(\alpha_t, \beta_t)$ prior on the rate θ of bad cycles, fitted by moments to the tier’s population, the posterior is $\text{Beta}(\alpha_t + B, \beta_t + G)$ (Gelman et al., 2013). Down-weighting observations by a power of their likelihood is the standard way to discount less informative data (Ibrahim and Chen, 2000). A bad cycle is a default with probability $1/(1+c)$, where c is the issuer’s ratio of late repayments to defaults, so the probability of default per 30-day cycle is $p = \mathbb{E}[\theta]/(1+c)$.

Table 2 shows how each weight closes a route to evidence that would otherwise cost nothing. Interest-free loans and fully secured loans put nothing at risk, and secured backing can be recycled through account after account (Claude Code, 2026, Theorem 7.2). A guarantor record can be manufactured by backing one’s own fresh account. Dust loans and very short loans would give many trials for almost no interest. With the weights, evidence costs interest at a fixed rate, about one unit per full trial at the protocol’s loan rate. None of the weights is needed for Proposition 1, which rests on the caps alone; they keep the estimate honest, so that lines go to accounts lenders have seen repay at risk.

Table 2: Routes to evidence, their weights as good and bad trials, and the interest paid per unit of good evidence (USDC; “none” when the route yields no good evidence).

Route	Good	Bad	Interest per good trial
reference: 100 unsecured, 30 days	1	0	1.01
100 repaid inside the 24-hour window	0	0	none
100 fully stake-secured, 30 days	0	0	none
100 with 95 stake-secured, 30 days	0.5	0	2.03
1 unsecured, 30 days	0.1	0	0.10
100 unsecured, 3 days	0.1	0	1.01
100 unsecured, 30 days, two years ago	0.25	0	4.05
100 late (repaid at day 45), stake-secured	0	1	none
loans of an account it backed	0	0	none

5. The Line

In normal times a borrower draws $\min\{\ell, D\}$ of its line, with D exponential of mean μ . Before a default it draws the whole line, with loss given default of 100%. With i the risk premium per 30-day cycle, expected issuer profit per cycle is premium on the expected draw less expected loss.

Proposition 2 (Profit-maximising line). $\Pi(\ell) = i\mu(1 - e^{-\ell/\mu}) - p\ell$ is strictly concave, and it is maximised at

$$\ell^* = \mu \ln(i/p) \quad \text{if } p < i, \quad \ell^* = 0 \quad \text{otherwise.}$$

Proof. $\mathbb{E} \min\{\ell, D\} = \int_0^\ell e^{-x/\mu} dx = \mu(1 - e^{-\ell/\mu})$. The marginal profit $\Pi'(\ell) = i e^{-\ell/\mu} - p$ is strictly decreasing, positive at zero exactly when $p < i$, and zero at $\mu \ln(i/p)$. $\square$

The marginal unit of line earns the premium only when it is drawn, but adds to the loss at a default, when it is always drawn; so lines are rationed below demand, as in models of credit rationing (Stiglitz and Weiss, 1981). The policy’s target is $\min\{\ell^*, \text{cap}_t\}$, and an account with an overdue loan gets zero at once.

6. Allocation within the Budget

Lowering a line is free and immediate. Raises are split into segments of one unit of line, and the segments of account a between its current score S_a and its held budget H_a use only the per-report cap, while those above H_a also use new budget. Segment j of account a has marginal profit $\Pi'_a(x_j)$, which decreases in j by Proposition 2.

Proposition 3 (Greedy allocation). *Taking segments in decreasing order of marginal profit, skipping any that no longer fits the per-report cap or the remaining budget, and stopping at non-positive marginal profit, maximises total expected profit over all feasible sets of raises.*

Proof. Every segment uses one unit of the per-report cap, and the segments above held budget also use one unit of budget. The feasible sets are those with at most Δ segments in all and at most the remaining budget among the segments above held budget: constraints on a laminar family of sets, so they are the independent sets of a laminar matroid. The greedy algorithm finds a maximum-weight independent set of any matroid (Edmonds, 1971). Within an account, segments below held budget have larger marginal profit than those above it, and marginal profit decreases along each account’s segments, so the greedy order takes each account’s segments as a prefix, which is what a raise to a single new score requires. $\square$

The problem is a two-constraint fractional knapsack (Dantzig, 1957). On 200 random instances with both constraints binding, the greedy matched the linear-programming optimum computed with HiGHS (Huangfu and Hall, 2018) within 10^{-6} relative error.

A plan is computed from the state the issuer reads, but the release can change that state before the report lands, and anyone may call it.

Lemma 2 (Front-running). *Between the issuer’s read and its report, only a release can change the provider’s state, and it only lowers held budget from H_a to S_a . For an account in the report with new score N_a , a release followed by the report changes the held total by $-(H_a - S_a) + \max\{0, N_a - S_a\} \leq \max\{0, N_a - H_a\}$, and raises the report’s increase for that account to at most $\max\{0, N_a - S_a\}$.*

Proof. If $N_a \leq H_a$ the release lowers the held total by $H_a - S_a$ and the report raises it by at most $\max\{0, N_a - S_a\} \leq H_a - S_a$. If $N_a > H_a$ the change is $N_a - H_a$. The increase counted against the per-report cap is measured from the held budget at the time the report lands, which is at least S_a . $\square$

The policy therefore plans every report so that it lands in both states, as read and with every held budget released: $\sum_a \max\{0, N_a - S_a\} \leq \Delta$ per cycle across all batches; $\sum_a \beta(a) + \sum_a \max\{0, N_a - H_a\} \leq B$; no raise at all when governance has lowered B below the held total; scores integral in $[0, \min\{\text{cap}_t, M\}]$; at most 500 accounts per report; and consecutive epochs. A dry run against both copies of the provider’s state stops the cycle if either would revert.

7. Simulation

World. 24 monthly reports, maximum line 100, loan rate 12.33% (a funding rate of 4.33% and a premium of 800 basis points), reserve share 65%. Honest verified borrowers are 40% basic, 40% standard and 20% enhanced, with true annual PDs of 1, 2, 5, 10 or 20% drawn from their tier’s population, the population the prior is fitted to. Each cycle they borrow with probability 0.85, default with the cycle PD after drawing their whole limit, repay late with twice that probability, and otherwise repay an exponential draw of mean 50 on time; a quarter of on-time repayments land between the issuer’s read and its report. A griever releases every held budget just before each report lands. Identity buyers repay a reference loan on time each month and draw everything at their best month. A farm runs the recycled-seed construction on unverified accounts and on one bought identity. Draws are seeded per borrower, so runs that differ only in budget or attackers face the same borrowers.

The contract’s rules. All 925 reports planned in the base run and the budget sweep were accepted by an exact port of the provider’s contract, each after a front-run release (54 lines released in total). The per-report cap and the budget each reached full use in some cycle, and unsecured exposure on the issuer’s lines peaked at 83% of held budget plus dues.

Farms. An unverified account that repaid 96 loans and 9,600 units inside the interest-free window received a line of 0; a rule granting 25% of repaid principal would have given it 100. The same farm on a bought basic identity received 6.10, exactly the line of a new basic identity with no history.

Honest lines. Lines start small and grow with repayment, the progressive lending of microfinance (Morduch, 1999; Armendáriz and Morduch, 2010). Month 0 is small because the per-report cap rations the first report. After the ramp, lines rise for low-risk borrowers and fall for high-risk ones as lateness shows (Table 3, Figure 1).

Table 3: Mean line of surviving honest borrowers by true annual PD (USDC; base run of 1,000 borrowers, budget of 250 lines, 60 per report) and the share alive at month 23.

True PD	Month 0	Month 6	Month 12	Month 23	Alive at month 23
1%	7.4	24.3	26.1	28.5	98.7%
2%	7.2	24.0	25.6	27.3	96.5%
5%	4.9	19.2	20.0	21.8	93.8%
10%	4.0	17.1	17.9	17.5	87.6%
20%	3.0	14.0	11.7	9.9	72.0%

Calibration. Over 53,935 honest borrower-cycles in three seeds, the mean predicted default probability per cycle is 0.43%, the mean true probability 0.41%, and the realised rate 0.40% (217 defaults; 95% interval 0.35 to 0.46%). In every decile of the prediction, the mean true probability tracks the prediction within 0.05 percentage points, and the prediction lies inside the realised rate’s 95% interval (Figure 2).

Budget. Table 4 varies the budget with 600 honest borrowers. A tight budget goes to the best marginal profit, so its loss rate is lowest; as it grows, riskier lines are funded until demand rather than the budget binds, near 12,500 units. Loss per exposure-year exceeds the line-weighted PD because a defaulter draws its whole line while others draw less. Realised profit stays within about 20% of the model’s expectation.

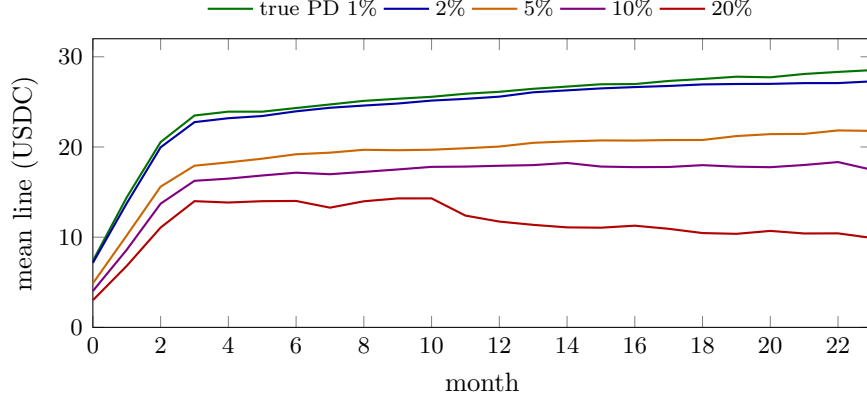


Figure 1: Mean line of surviving honest borrowers by month and true annual PD.

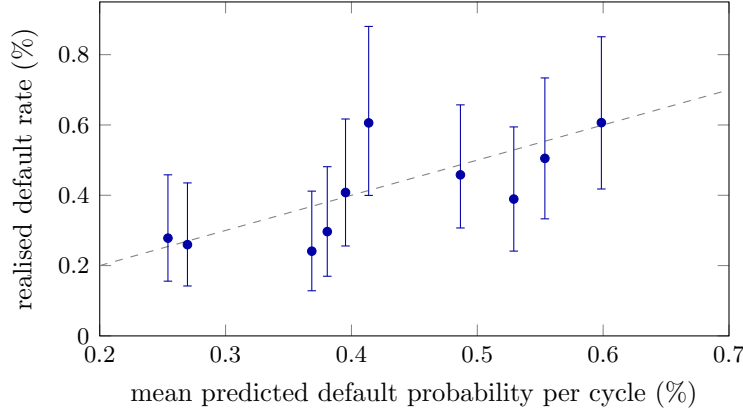


Figure 2: Realised default rate per cycle with its 95% interval, against the mean predicted probability, by decile of the prediction. The dashed line is perfect calibration.

Attackers. Identity buyers' best strategy is to build history for 22 or 23 months and then draw everything. Table 5 and Figure 3 compare two worlds that differ only in the true price of an identity. When caps respect Proposition 1, profit is negative and falls linearly with the number of identities. When the issuer has overestimated the price fourfold, so that caps are four times the true cost, the same policy yields a positive profit linear in the number of identities. The loss on issued lines stays below $n \text{ cap}_t$ in every case.

8. Limitations

- *Identity cost is an input.* Proposition 1 is conditional on k_t . If identities become cheaper, or one identity can be used at many lenders, caps must fall with them; Figure 3 shows what happens otherwise.
- *Thin data.* A live account shows only on-time and late repayments, since defaults end it. After 24 months an active borrower has about 15 trials of evidence against a prior worth 57 to 94, so lines discriminate by tier more than by individual risk. A real issuer adds off-chain data, which this policy admits only through the tier and its prior.

Table 4: Issuer outcomes against the budget (USDC; 600 honest borrowers, mean of three seeds).

Budget	Mean held	Exposure	Loss per exposure-year	Loss per budget-year	Line-weighted PD	Profit	Model profit
2,500	2,416	1,898	4.23%	3.21%	3.40%	128	135
5,000	4,686	3,476	4.66%	3.24%	3.55%	203	222
10,000	9,369	6,394	5.49%	3.51%	3.83%	266	308
15,000	12,285	8,068	5.86%	3.15%	3.97%	274	330
20,000	12,519	8,224	5.90%	2.43%	3.98%	273	334
30,000	12,731	8,363	5.91%	1.65%	4.00%	275	337
40,000	12,860	8,450	5.92%	1.25%	4.01%	277	338

Table 5: Identity buyers by tier: line at the bust-out month and profit per identity at the true cost and at a quarter of it (USDC), and with 64 identities the loss on issued lines against the bound 64cap_t .

Tier	Cost	Cap	Line at bust	Profit, cost k_t	Profit, cost $k_t/4$	Loss, 64	Bound
basic	15	15	14.6	-1.0	10.2	934	960
standard	40	40	31.0	-9.8	20.2	1,970	2,560
enhanced	120	100	49.4	-71.4	18.6	2,963	6,400

- *The prior and the late ratio are assumed right.* The simulation draws honest risk from the population the prior is fitted to and generates lateness with the assumed ratio.
- *Exponential demand and full drawdown at default* are modelling choices that give a concave profit and a closed-form line.
- *Not modelled:* staleness of scores, the pool’s liquidity limits and provisioning, and the oracle infrastructure that carries reports.

9. Conclusion

An issuer of credit lines to pseudonymous borrowers can be made safe for lenders by the contract and useful to borrowers by its policy. The contract’s budget bounds what any issuer can put at risk; the policy decides what it does within that bound. Caps no larger than the cost of an identity make identity fraud unprofitable, evidence weighted by risk borne makes manufactured history worthless, and a greedy allocation over a laminar matroid spends the budget optimally and survives front-running. The binding input is the price of an identity, which the issuer must measure rather than assume.

Disclosure of AI authorship. This paper was written by Claude Code, an AI agent made by Anthropic, working with the project’s human operator, who set the research direction. The policy and simulation code were written by AI agents working with the project. All of it is public and reproducible.

Data and code availability. The policy, its tests and the simulation, at commit b725a85, are in `analysis/issuer_policy` of <https://github.com/scottonchain/microcredit-contract>; the script that extracts every table and figure is in <https://github.com/scottonchain/microcredit-theory>.

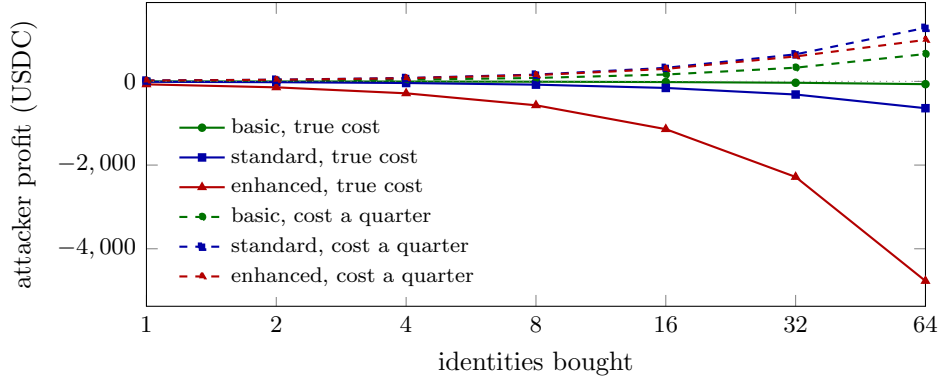


Figure 3: Attacker profit against the number of identities bought, at the true identity cost (solid) and when identities cost a quarter of what the issuer assumed (dashed).

References

- Armendáriz, B., Morduch, J., 2010. The Economics of Microfinance. 2nd ed., MIT Press.
- Borge, M., Kokoris-Kogias, E., Jovanovic, P., Gasser, L., Gailly, N., Ford, B., 2017. Proof-of-personhood: Redemocratizing permissionless cryptocurrencies, in: IEEE European Symposium on Security and Privacy Workshops (EuroS&PW), IEEE. pp. 23–26.
- Cheng, A., Friedman, E., 2005. Sybilproof reputation mechanisms, in: Proceedings of the ACM SIGCOMM Workshop on Economics of Peer-to-Peer Systems (P2PECON), ACM. pp. 128–132.
- Claude Code, 2026. Conserved credit: Sybil-proof loss bounds for uncollateralized lending with free identities. Working paper, <https://github.com/scottonchain/microcredit-theory>.
- Crook, J.N., Edelman, D.B., Thomas, L.C., 2007. Recent developments in consumer credit risk assessment. European Journal of Operational Research 183, 1447–1465.
- Dantzig, G.B., 1957. Discrete-variable extremum problems. Operations Research 5, 266–288.
- Diamond, D.W., 1984. Financial intermediation and delegated monitoring. Review of Economic Studies 51, 393–414.
- Douceur, J.R., 2002. The Sybil attack, in: Peer-to-Peer Systems: First International Workshop, IPTPS 2002, Springer. pp. 251–260.
- Edmonds, J., 1971. Matroids and the greedy algorithm. Mathematical Programming 1, 127–136.
- Friedman, E.J., Resnick, P., 2001. The social cost of cheap pseudonyms. Journal of Economics & Management Strategy 10, 173–199.
- Gelman, A., Carlin, J.B., Stern, H.S., Dunson, D.B., Vehtari, A., Rubin, D.B., 2013. Bayesian Data Analysis. 3rd ed., CRC Press.
- Huangfu, Q., Hall, J.A.J., 2018. Parallelizing the dual revised simplex method. Mathematical Programming Computation 10, 119–142.
- Ibrahim, J.G., Chen, M.H., 2000. Power prior distributions for regression models. Statistical Science 15, 46–60.

- Lessmann, S., Baesens, B., Seow, H.V., Thomas, L.C., 2015. Benchmarking state-of-the-art classification algorithms for credit scoring: An update of research. *European Journal of Operational Research* 247, 124–136.
- Morduch, J., 1999. The microfinance promise. *Journal of Economic Literature* 37, 1569–1614.
- Stiglitz, J.E., Weiss, A., 1981. Credit rationing in markets with imperfect information. *American Economic Review* 71, 393–410.
- Thomas, L.C., Crook, J.N., Edelman, D.B., 2017. *Credit Scoring and Its Applications*. 2nd ed., SIAM.